

Hyunjoon Lee

Seoul National University, Seoul 08826, Republic of Korea

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EDUCATION

- **Integrated M.S./Ph.D. in Artificial Intelligence**

Seoul National University

◦ Advisor: Jaesik Park

Mar. 2024 – Present

Seoul, Korea

- **B.E. in Mechanical Engineering**

Sungkyunkwan University

◦ GPA: 4.39/4.5

◦ Summa Cum Laude (1/131)

Mar. 2018 – Feb. 2024

Suwon, Korea

RESEARCH INTERESTS

- Vision-Language-Action Model / World Action Model • Harness • 3D Vision

RESEARCH IN PROGRESS

- **Zero-shot History-Aware Grasping via Vision-Language Models**

Project Lead & First Author

- **Vision-Free In-Hand Dexterous Manipulation via Force Sensing**

Co-Author

- **Open-X-Tactile: Advancing Generalist Tactile Policy through a Unified Open Data Ecosystem**

Contributor

PUBLICATIONS

* denotes equal contribution

INTERNATIONAL

- **Relation-Centric Open-Vocabulary 3D Gaussian Segmentation**

Eunsung Cha*, Hyunjoon Lee*, Jaesik Park

Under Review, 2026

We propose a novel framework that reframes Gaussian segmentation as modeling pairwise relations between Gaussians, unifying instance-level coherence and query flexibility without per-scene optimization.

- **Occluded 3D Object Reconstruction via Masked Multi-view Volumetric Transformer**

Jinmo Kim, Namtae Kim, Hyunjoon Lee, Seungha Kim, Jaesik Park

Under Review, 2026

We propose a feed-forward multi-view 3D reconstruction framework that explicitly reasons about occlusions by reconstructing missing latent tokens across views.

- **CF3: Compact and Fast 3D Feature Fields**

Hyunjoon Lee, Joonkyu Min, Jaesik Park

Int. Conf. on Computer Vision (ICCV), Main Conference & Demonstrations Track, 2025

We propose a method to build compact and fast 3D Gaussian feature fields by effectively compressing and sparsifying Gaussians, achieving competitive performance with significantly fewer Gaussians.

DOMESTIC

- **Language-Driven Robotic Manipulation via Open-Vocabulary 3D Gaussian Scene Understanding**

Hyunjoon Lee, Eunsung Cha, Jaesik Park

KCC (Outstanding Paper Presentation Award), 2026

- **Aligning 2D Mask Space Consistency Using 3D Gaussian Splatting**

Eunsung Cha, Hyunjoon Lee, Jaesik Park

KCC (*Outstanding Paper Presentation Award*), 2026

- **Efficient Feature Lifting and Compression Using Pre-trained 3D Gaussians**

Hyunjoon Lee, Joonkyu Min, Jaesik Park

KCC (*Outstanding Paper Presentation Award*), 2025

PROJECTS

- **Real2Sim VLA**

Personal Project

Fine-tuned a VLA on real-world scene data and reconstructed the same scene with 2DGS into a USD-compatible mesh for Isaac Sim, enabling photorealistic Real2Sim post-training and scene-level evaluation.

- **RAG-based Interactive 3D Car Live Video Service using CF3**

Kolon Motors, Korea, Sep. 2025 – Dec. 2025

- **End-to-End 3D Reconstruction of Complex Scenes for Robotic Perception**

Hyundai Motors, Korea, Mar. 2025 – Mar. 2026

- **Robust Pick and Place**

LG Electronics, Korea, Jun. 2023 – Sep. 2023

- **Radio Station Protection and Frequency Sharing via Spectrum Challenge**

Institute of Information & Communications Technology Planning & Evaluation, Korea, Jan. 2023 – Jun. 2023

- **Design of a Two-Stage Reducer**

SKKU, Korea, Mar. 2020 – Jun. 2020

EXPERIENCES

- **Research intern**

HuGe Lab, KAIST AI

Jun. 2023 – Sep. 2023

Seoul, Korea

◦ *Advisor: Beomjoon Kim*

◦ Developed a model to predict 6 DoF grasp poses using a depth sensor

- **Research intern**

CSI Lab, SKKU

Jan. 2023 – Jun. 2023

Suwon, Korea

◦ *Advisor: Yusung Kim*

◦ Developed an RL agent using the PPO algorithm and transformer architecture

- **Research intern**

CAMAS Lab, SKKU

Dec. 2019 – Mar. 2020

Suwon, Korea

◦ *Advisor: Jachoon Koo*

◦ Developed a mathematical model to elucidate the stiffness mechanism of the soft gripper

AWARDS AND HONORS

- **Outstanding Paper Presentation Award**, “Language-Driven Robotic Manipulation via Open-Vocabulary 3D Gaussian Scene Understanding”, Korea Computer Congress, Korea (Aug. 2026)

- **Outstanding Paper Presentation Award**, “Aligning 2D Mask Space Consistency Using 3D Gaussian Splatting”, Korea Computer Congress, Korea (Aug. 2026)

- **Outstanding Paper Presentation Award**, Workshop on Image Processing and Image Understanding (IPIU), Korea (Feb. 2026)

- **Honorable Mention**, Research Paper Competition, Seoul National University, Korea (Dec. 2025)

- **Outstanding Paper Presentation Award**, “Efficient Feature Lifting and Compression Using Pre-trained 3D Gaussians”, Korea Computer Congress, Korea (Aug. 2025)

- **Invited Talks**, “CF3: Compact and Fast 3D Feature Fields” SNU Graduate of AI Research Exchange, Seoul National University, Korea (Jun. 2025)
- **Top Excellence Award at the Creative Research Project**, Seoul National University, Korea (Jun. 2025)
- **Outstanding Award at the demo competition**, Seoul National University, Korea (Jun. 2025)
- **Invited Talks**, “CF3: Compact and Fast 3D Feature Fields”, NaverLabs, Korea (Mar. 2025)
- **Dean’s List Award**, SungKyunKwan University(SKKU), Korea (May. 2021)
- **Scholarship for academic excellent**, SungKyunKwan University(SKKU), Korea (Mar. 2021)
- **Dean’s List Award**, SungKyunKwan University(SKKU), Korea (Oct. 2020)
- **Scholarship for research excellent**, SungKyunKwan University(SKKU), Korea (Mar. 2020)
- **Scholarship for academic excellent**, SungKyunKwan University(SKKU), Korea (Mar. 2019)
- **Scholarship for academic excellent**, SungKyunKwan University(SKKU), Korea (Sep. 2018)

PATENT

- CF3: Compact and Fast 3D Feature Fields, 10-2025-0093174

TEACHER ASSISTANT

- **Samsung Electronics DS Cooperation Course**, Samsung Electornics (Aug. 2025)
- **Introduction to Artificial Intelligence**, Seoul National University (2025 Spring)
- **Samsung Electronics DS Cooperation Course**, Samsung Electornics (Aug. 2024 – Oct. 2024)
- **Introduction to Artificial Intelligence**, Seoul National University (2024 Spring)